SWARRNIM INSTITUTE OF TECHNOLOGY

B.Tech 1st SEMESTER — PYTHON PROGRAMMING MOCK TEST (SET 1)

Practice Paper | Similar Pattern and Difficulty, Different Questions

Instructions:

Attempt all questions in Q.1.

In Q.2, Q.3 and Q.4, attempt any TWO sub-questions from each section.

Each question in Q.1 (A) and Q.1 (B) carries 1 mark. Each attempted sub-question in Q.2, Q.3 and Q.4 carries 5 marks.

Use any appropriate Python approach. No specific loop or conditional method is required unless stated in a question.

Q.1 (A) Multiple Choice Questions (MCQs) — 05 Marks

Attempt all questions. Each question carries 1 mark.

1. What is the value of the expression?
print(18 // 4 + 18 % 4)

A. 4 B. 6

C. 2 D. 8

2. What is printed?
a = 5
b = 2
a *= b + 1
print(a)

A. 12 B. 15

C. 10 D. 7

3. What is the output?
s = "PYTHON"
print(s[-4:-1])

A. YTH B. THO

C. YTHO D. HON

4. What is printed?
print(0 or 8 + 5 or 0)

A. 0 B. 13

C. 5 D. 8

5. What is printed?
x = 7
if x % 2 == 0:
print('A')
else:
print('B')

A. A B. B

C. 7 D. No output

Q.1 (B) Short Questions (Define / Explain) — 05 Marks

1. What is the difference between a syntax error and a logical error?

2. What does bool() do? Give one example.

3. What is the purpose of the comparison operators == and !=?

4. What is string slicing? Explain the role of the start and stop positions.

5. What is the purpose of the range() function in a loop?

Q.2 — Attempt Any TWO — 10 Marks

(A) Explain comparison operators and logical operators. Write one example expression for each category and state its result.

(B) Explain the difference between a for loop and a while loop. Give a small example of each.

(C) Explain nested if statements and show a program that checks whether a person is eligible for a student discount based on age and student status.

Q.3 — Attempt Any TWO — 10 Marks

(A) Electricity Bill Category Write a program that accepts units consumed and prints the category: below 100 → Low Usage; 100 to 299 → Medium Usage; 300 or more → High Usage. Assume the input is non-negative.
Test cases: 75 → Low Usage; 180 → Medium Usage; 350 → High Usage.

(B) Prime Number Check Write a program that accepts an integer greater than 1 and determines whether it is prime or not prime.
Test cases: 11 → Prime; 15 → Not Prime; 2 → Prime.

(C) Number Pattern Write a program that accepts N and prints the following pattern.
Input: 5
Expected Output:
54321
4321
321
21
1

Q.4 — Attempt Any TWO — 10 Marks

(A) Count Vowels Write a program that accepts a word and counts how many vowels it contains. Treat uppercase and lowercase vowels alike.
Test cases: education → 5; SKY → 0; Apple → 2.

(B) Multiplication Table Range Write a program that accepts a number and prints its multiplication table from 1 to 10.
Test cases: N = 3 → 3 6 9 12 15 18 21 24 27 30.

(C) Nested Loop Output Predict the output, then verify it by running the code:

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for i in range(1, 4):
for j in range(i):
print(j + 1, end='')
print()
```

Also explain how many times the inner loop runs in total.

— End of Mock Test —